Spectroscopy

Pupil sheet

Name: _____ **Date:** _____

Safety: The box is not waterproof: keep water away from it. Goggles with dyes; taste nothing.

1. What I see on the screen

Only Intensity I (lx) is measured; **Transmittance** (%) and **Absorbance** are worked out against the reference set with **Set I_0 (reference)**, with the blank (the cup with clear water) in the beam.

The rule that saves the series: if the app asks me to set the reference again ($I > I_0$), the lamp has changed, something has moved, or room light has crept in. I reset it and start again.

I_0 = _____ **lx**
Write it here: it does *not* appear in the exported file, and without it you cannot recompute anything at home.

2. What I do on my own

First I predict. I do not change my prediction after I measure.
If I dilute the solution by half, the absorbance will ☐ halve ☐ fall by less than half ☐ stay the same

because _____

_____

The filters. Between filters I return to a clear path and check that the reading comes back to I_0. If it does not, something has moved.

Filter	Light (lx)	T (%)	A
Two stacked:			

I check the logarithm: $T_1 \times T_2$ = _____ %, and $A_1 + A_2$ = _____. Do they match the pair I measured? ☐ yes ☐ no

The dilution series. The volumes are already written in. **The reference is set with the water blank in the slot**, not with an empty slot. The control row in the middle is a blank too: if it no longer gives close to zero, the series went wrong between it and the start.

Concentration	Stock	Water	Light (lx)	A
Blank (water)	0 mL	10 mL		
20%	2 mL	8 mL		
40%	4 mL	6 mL		
Blank (control)	0 mL	10 mL		
60%	6 mL	4 mL		
80%	8 mL	2 mL		
100%	10 mL	0 mL		
Unknown	—	—		

Lamp: _____ **Lamp-to-sensor distance:** _____ cm **Room lights:** ☐ off

3. The conclusion I write

I predicted that, and the data show that _____

_____

My line was straight up to $A =$ _____, and after that it bent because _____

_____

The concentration of the unknown, read off the line, is _____

_____

Exit ticket

Why does an absorbance mean nothing without a reference?

_____